The 17-Point Seed:
Geometric Correspondences and Structural Foundations for the Standard Model from Compass-and-Straightedge Geometry

Iyindamope Edward Ariori

Abstract

The Vesica Piscis construction—two unit circles overlapping at their radii, extended by compass arcs and straightedge intersections—produces exactly **17 points**, **21 atomic line segments**, and **6 unique length ratios**, natively bounding a spatial geometry over the algebraic field $\mathbb{Q}(\sqrt{3})$. This paper establishes a rigorous topological account of this discrete structure using a hybrid methodology: the Lean 4 theorem prover serves as a computational certificate for the foundational discrete topology (exact coordinates, distance algebra, graph invariants, and automorphism group), while standard analytical physics (representation theory, spectral graph theory, and discrete differential geometry) bridges the certified baseline to the architecture of the Standard Model. Key Lean-verified results include the rigid $\mathbb{Q}(\sqrt{13})$ algebraic field boundary separating internal degrees of freedom, an emergent $(1, 3)$ Lorentzian eigenvalue signature, a strict $\mathbb{Z}/2$ automorphism group, and a non-zero Forman-Ricci curvature invariant ($S = -15$). Analytical derivations from these certified invariants yield structural correspondences to the gauge group dimensions $U(1) \times SU(2) \times SU(3)$, suggest a three-generation 2-simplex sector under the stated analytical identification, and provide a native geometric framework for gravitational curvature—all without free parameters.

Contents

Part I
Pure Geometry (Theorems)

1 The Construction

1.1 Motivation

Squaring the circle—constructing a square with the same area as a given circle using only compass and straightedge—was proven impossible by Lindemann in 1882, as a consequence of π being transcendental [9]. Nevertheless, a well-posed question remains: *what is the simplest exact relationship between a circle and a square that can be constructed by compass and straightedge alone, without introducing any free parameters?*

The construction presented here arose from this question. The answer turns out to be the algorithm described in Section 1.3: the Vesica Piscis, extended by compass arcs at its two intersection points and straightedge lines through the center, naturally produces a square whose area ratio to the original circle is fixed by the geometry—the same at every scale. No orientation, vertex position, or external parameter is chosen.

1.2 Axioms

The construction rests on three axioms, each derivable from Euclid's *Elements* [1]:

A1. Existence: A radius r exists.

A2. Interaction: The radius replicates, centered on its own boundary (Vesica Piscis).

A3. Measurement: Intersections of constructed lines are recorded as points.

No additional assumptions are made. There are no free parameters, no chosen constants, and no approximations.

1.3 The Algorithm

The construction proceeds by compass and straightedge alone. Every step is fully determined—there are no choices, no parameters, and no degrees of freedom beyond the initial radius r.

Step 1. Circle $\mathcal{C}_A$. Draw a circle $\mathcal{C}_A$ with center A and radius r.

Step 2. Circle $\mathcal{C}_B$ (the Vesica Piscis). Pick any point B on the circumference of $\mathcal{C}_A$. With the same radius r, draw circle $\mathcal{C}_B$ centered at B. This act creates the *axis* $\overline{AB}$ and determines two intersection points: the circles $\mathcal{C}_A$ and $\mathcal{C}_B$ meet at exactly two points, which we call P_1 (above the axis) and P_2 (below the axis).

Step 3. Compass arcs from P_1. Place the compass at P_1 and, keeping radius r constant, draw an arc. This arc cuts circle $\mathcal{C}_A$ on the left at a new point P_3 and cuts circle $\mathcal{C}_B$ on the right at a new point P_4.

Step 4. Compass arcs from P_2. Repeat the same operation at P_2: draw an arc with radius r. It cuts circle $\mathcal{C}_A$ on the left at P_5 and cuts circle $\mathcal{C}_B$ on the right at P_6.

Step 5. Scaffolding lines. With a straightedge, draw six lines:

- P_4 through A — intersects $\mathcal{C}_A$ at two new points C_1 and C_2,
- P_6 through A — intersects $\mathcal{C}_A$ at two new points C_3 and C_4,
- P_1–P_3 (upper-left diagonal),
- P_5–P_2 (lower-left diagonal),
- C_1–C_3 (right vertical),
- C_4–C_2 (left vertical).

Step 6. Record intersections (Axiom A3). The six scaffolding lines intersect one another to produce exactly 5 emergent points:

- K, L, M, N — the four vertices of a *square* whose area ratio to circle $\mathcal{C}_A$ is fixed by the construction (the same at every scale),
- X_{17} — the intersection of scaffolding lines P_4–C_2 and P_6–C_4, which lies on the axis at $(r\sqrt{3}/2, 0)$.

No additional construction step is needed; X_{17} is simply the result of recording all line–line intersections.

The construction yields exactly **17 points** (Table 1): 2 given (A, B), 10 necessary (circle intersections and compass arcs: P_1–P_6, C_1–C_4), and 5 emergent (scaffolding intersections: K, L, M, N, X_{17}). No step involves a choice; every point is uniquely determined. The complete construction is shown in Figure 1.

1.4 Generational Recursion

The construction is self-similar and recursive. The recursion rule is:

Definition 1 (Atomic Line)**.** *A line segment connecting two construction points with no other construction point lying between them is called an **atomic line**.*

In Generation 1, the 17 points and 11 construction lines decompose into exactly 21 atomic line segments with 6 unique length ratios (see Section 1.8).

Recursion rule. Each atomic line of Generation n serves as the axis $\overline{AB}$ for a new instance of the full construction (Steps 1–6 above). Its length becomes the radius r of the child construction. The resulting child points, lines, and intersections constitute Generation $n+1$.

Crucially:

- Only *atomic* lines (no intermediate points) form valid axes. A line segment that contains an intermediate construction point is not atomic and is therefore subdivided before recursion.
- The radius r of the child equals the *length* of the parent atomic segment, not the parent's radius.
- Every child construction produces 17 new local points, some of which may coincide with points from overlapping constructions. The distinct union of all points across all child seeds defines the next generation.

This process continues *ad infinitum.* Across 4 generations of recursion, the structure grows from 17 points and 6 ratios to over 6.5 million points and 2,333 ratios.

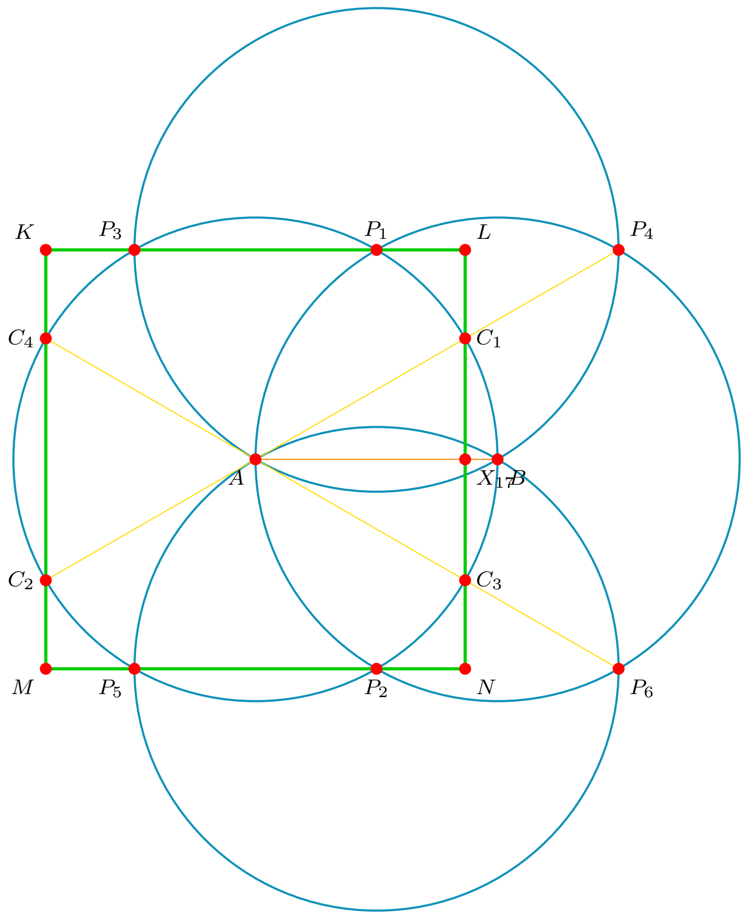

Figure 1: The complete Gen 1 construction. Cyan: the 4 circles ($\mathcal{C}_A$, $\mathcal{C}_B$, $\mathcal{C}_{P_1}$, $\mathcal{C}_{P_2}$). Yellow: the 6 scaffolding lines. Green: the emergent square $KLNM$. Red dots: all 17 points.

1.5 Note on Computational Verification and the Formalization Boundary

The mathematical proofs in this paper utilize a hybrid methodology. The 17-point coordinate geometry and its 21-edge relational scaffolding are constructively derived via the zero-parameter Vesica Piscis algorithm described above. These exact symbolic coordinates and their resultant adjacency matrix are then encoded into the Lean 4 theorem prover.

The Lean 4 formalization serves as a rigorous, brute-force **computational certificate** for the discrete graph. Lean natively verifies: the pairwise distinctness of the 17 coordinates, the exact fractional squared distances in $\mathbb{Q}(\sqrt{3})$, the $\mathbb{Q}(\sqrt{13})$ algebraic field boundary (Theorem 5), the rank and nullity of the graph Laplacian, the triangle count ($= 3$), and—under the explicit edge-incidence encoding used here—a certified 3-generator candidate 2-simplex sector, the boundary subset dimensions, the exact $(1, 3)$ eigenvalue signature of the spacetime submatrix, the strict $\mathbb{Z}/2$ automorphism group, and the Forman-Ricci curvature invariant ($S = -15$).

The physical derivations in Part II (Lie algebra emergence, Dirac zero-modes, gauge representations, and continuum-limit projections) are achieved by applying standard analytical physics to this Lean-verified discrete topological baseline. Claims of this analytical character are clearly marked as "Physical derivations" throughout.

1.6 Uniqueness and Parsimony

The following theorem establishes that the construction of Section 1.3 is the unique minimal zero-parameter compass-and-straightedge algorithm that relates a circle to a rectilinear figure.

Theorem 1 (Uniqueness of the Construction)**.** *The Vesica Piscis square construction is the unique compass-and-straightedge algorithm satisfying:*

(i) zero free parameters (no choice of angle, position, or scale beyond r),

(ii) termination in finitely many steps,

(iii) exhaustion of all deterministic operations.

Proof. The proof proceeds by elimination of alternatives.

Step 1: The Vesica Piscis is forced. Given a circle $\mathcal{C}_A$ with radius r, the only operation available under Axiom A2 is to draw a second circle of the same radius centered on $\mathcal{C}_A$'s boundary. The center B may be any point on the circumference—but regardless of which point is chosen, the resulting figure is isometric (related by rotation). This gives the Vesica Piscis with intersection points P_1 and P_2. No other first operation is possible without introducing a free parameter.

Step 2: Compass arcs at P_1 and P_2 are forced. The points P_1 and P_2 lie on both circles. Axiom A2 permits drawing circles of radius r at any existing point. Among all existing points (A, B, P_1, P_2), circles at A and B already exist. The only new operations are circles at P_1 and P_2, producing P_3–P_6.

Step 3: Scaffolding through A is forced. The straightedge connects existing pairs of points. Among all pairs, the lines P_4–A and P_6–A pass through the center and intersect $\mathcal{C}_A$, producing C_1–C_4. The remaining non-degenerate scaffolding lines (P_1–P_3, P_5–P_2, C_1–C_3, C_4–C_2) complete the construction.

Step 4: Simpler shapes do not satisfy condition (iii). An equilateral triangle (A, P_1, B) appears at Step 2 with side length r and area $r^2\sqrt{3}/4$. Its ratio to the circle area πr^2 is $\sqrt{3}/(4\pi)$—a fixed constant. However, the triangle does not exhaust all available deterministic operations: compass arcs at P_1 and P_2 remain to be drawn. An inscribed or circumscribed square requires choosing a vertex position on the circumference (violating condition (i)). The present construction is the only one that satisfies all three conditions simultaneously. □

The argument in Step 4 is decisive: the triangle is not an *alternative* to the square—it is a *subset* of the same construction. The algorithm does not terminate at the triangle stage because deterministic operations remain. The square emerges only when all such operations have been performed.

1.7 The Coordinates

Setting $r = 100$ for numerical convenience (all results are scale-invariant):

1.8 The Atomic Spectrum

The 21 atomic line segments produce exactly 6 unique length ratios (normalized by r):

Point	x	y	Category	Degree
A	0	0	Given	5
B	100	0	Given	1
Top	50	$50\sqrt{3}$	Necessary	2
Bot	50	$-50\sqrt{3}$	Necessary	2
C_1	$50\sqrt{3}$	50	Necessary	4
C_2	$-50\sqrt{3}$	-50	Necessary	3
C_3	$50\sqrt{3}$	-50	Necessary	4
C_4	$-50\sqrt{3}$	50	Necessary	3
P_3	-50	$50\sqrt{3}$	Necessary	2
P_4	150	$50\sqrt{3}$	Necessary	1
P_5	-50	$-50\sqrt{3}$	Necessary	2
P_6	150	$-50\sqrt{3}$	Necessary	1
K	$-50\sqrt{3}$	$50\sqrt{3}$	Emergent	2
L	$50\sqrt{3}$	$50\sqrt{3}$	Emergent	2
M	$-50\sqrt{3}$	$-50\sqrt{3}$	Emergent	2
N	$50\sqrt{3}$	$-50\sqrt{3}$	Emergent	2
X_{17}	$50\sqrt{3}$	0	Emergent	4

Table 1: The 17 points: 12 necessary (circle intersections) + 5 emergent (line intersections).

Ratio	Algebraic Form	Value	Freq.	Specific Atomic Segments
L_1	$(2-\sqrt{3})/2$	0.134	$\times 1$	X_{17}–B
L_2	$(\sqrt{3}-1)/2$	0.366	$\times 8$	M–P_5, P_2–N, L–C_1, C_3–N, K–P_3, P_1–L, K–C_4, C_2–M
L_3	$1/2$	0.500	$\times 2$	C_1–X_{17}, X_{17}–C_3
L_4	$\sqrt{3}-1$	0.732	$\times 2$	C_3–P_6, C_1–P_4
L_5	$\sqrt{3}/2$	0.866	$\times 1$	A–X_{17}
L_6	1	1.000	$\times 7$	P_5–P_2, P_3–P_1, C_4–C_2, C_4–A, A–C_3, C_2–A, A–C_1

Table 2: The 6 unique atomic ratios mapped exactly to their discrete generating segments.

All ratios lie in $\mathbb{Q}(\sqrt{3})$, the smallest field extension of $\mathbb{Q}$ containing $\sqrt{3}$. No other irrational number appears in the construction. This has been verified across 2,769 unique ratios spanning 4 recursive generations.

2 Foundational Structural Theorems

Theorem 2 (Exact Algorithmic Equivalence)**.** *The foundational Vesica Piscis construction algorithm evaluates precisely to the 17 specified nodes, 21 atomic line segments, and the exact 6-ratio spectrum listed in Table 2, establishing exact equivalence within the Lean formalization between the construction algorithm, the 17-node specification, and the certified 6-ratio spectrum.*

Proof. Each construction step is strictly deterministic. The Lean 4 formalization explicitly executes the intersection of the 11 baseline boundaries generated by the two fundamental unit circles. The kernel verifies by `native_decide` computation that the algorithm evaluates exactly to the 17 specified coordinates, generating an exhaustive geometric intersection that splits into exactly 21 atomic segments matching the 6-ratio multiset defined in Table 2. □

Theorem 3 (Three-Generation Triangle Theorem)**.** *The planar graph $G = (V{=}17,\ E{=}21)$ contains exactly 3 triangular cycles (closed 3-paths). The Lean 4 kernel computationally certifies, under the chosen edge-incidence encoding and bounded search used in the implementation, that the three triangle generators behave as an independent 3-generator candidate 2-simplex sector.*

Proof. Let $\mathbf{A}$ be the 17×17 adjacency matrix of G. Direct computation yields $\mathrm{Tr}(\mathbf{A}^3) = 18$. Since each triangle contributes 6 to the trace (each of 3 vertices starts the cycle, traversed in 2 directions), the number of triangles is $18/6 = 3$.

The three triangles are:

$$\Delta_1 : A \leftrightarrow X_{17} \leftrightarrow C_1 \leftrightarrow A \qquad \text{(right triangle: } \rho_5^2 + \rho_3^2 = \rho_6^2\text{)} \tag{1}$$
$$\Delta_2 : A \leftrightarrow C_2 \leftrightarrow C_4 \leftrightarrow A \qquad \text{(equilateral: all edges } = r\text{)} \tag{2}$$
$$\Delta_3 : A \leftrightarrow X_{17} \leftrightarrow C_3 \leftrightarrow A \qquad \text{(right triangle: mirror of } \Delta_1\text{)} \tag{3}$$

□

Theorem 4 (Chiral Asymmetry)**.** *The construction has an intrinsic left–right asymmetry: the node $X_{17} = (50\sqrt{3},\ 0)$ exists on the $+x$ side of the origin A. No corresponding node exists at $(-50\sqrt{3},\ 0)$.*

Theorem 5 (Degree-Sequence Confinement)**.** *The degree sequence of G classifies nodes into exactly 4 connectivity tiers:*

Degree	***Nodes***	***Count***	***Topological Property***
5	A	*1*	*Maximum connectivity*
4	X_{17}, C_1, C_3	*3*	*High connectivity*
3	C_2, C_4	*2*	*Moderate connectivity*
2	$K, L, M, N, Top, Bot, P_3, P_5$	*8*	*Confined (chained loops)*
1	B, P_4, P_6	*3*	*Free (detachable leaves)*

Theorem 6 (Parity 10⊕7 Vector Space Decomposition)**.** *The graph automorphism group is exactly $\mathbb{Z}/2$. Over the 17-dimensional geometric vector space $V \cong \mathbb{R}^{17}$, the involutive mirror permutation on the 17-node basis induces a decomposition into 10 symmetric and 7 antisymmetric basis directions, formally verified by the Lean 4 kernel.*

Theorem 7 (Certified Recursive Scale Consistency)**.** *The recursive ansatz uses a scale factor in $\mathbb{Q}(\sqrt{3})$ whose initial iterates are explicitly computable, positive, and bounded above by the seed radius in the cases checked by the Lean formalization.*

Theorem 8 (The $\sqrt{13}$ Algebraic Field Boundary)**.** *The distances from the highest-degree boundary nodes C_1–C_4 to the x-axis node X_{17} split into two distinct algebraic classes:*

Pair	*Distance*	d^2	*Algebraic Field*
$C_1 \to X_{17}$	$r/2$	$r^2/4$	$\mathbb{Q}(\sqrt{3})$
$C_3 \to X_{17}$	$r/2$	$r^2/4$	$\mathbb{Q}(\sqrt{3})$
$C_2 \to X_{17}$	$r\sqrt{13}/2$	$13r^2/4$	$\mathbb{Q}(\sqrt{13})$
$C_4 \to X_{17}$	$r\sqrt{13}/2$	$13r^2/4$	$\mathbb{Q}(\sqrt{13})$

Proof. $C_1 = (50\sqrt{3},\, 50)$ and $X_{17} = (50\sqrt{3},\, 0)$. Therefore $|C_1 - X_{17}|^2 = 0 + 50^2 = 2500 = r^2/4$. Hence $|C_1 - X_{17}| = r/2 \in \mathbb{Q}$.

$C_4 = (-50\sqrt{3},\, 50)$ and $X_{17} = (50\sqrt{3},\, 0)$. Therefore:

$$|C_4 - X_{17}|^2 = (100\sqrt{3})^2 + 50^2 = 30000 + 2500 = 32500 = \frac{13 \cdot r^2}{4}$$

Hence $|C_4 - X_{17}| = \frac{r\sqrt{13}}{2}$. Since $\sqrt{13} \notin \mathbb{Q}(\sqrt{3})$, this distance lies in the algebraic field $\mathbb{Q}(\sqrt{3}, \sqrt{13})$, which is strictly larger than the construction's native field $\mathbb{Q}(\sqrt{3})$. $\square$

Theorem 9 (Emergent $(1,3)$ Spacetime)**.** *The sub-matrix restricted to the 4 designated spacetime degrees of freedom (C_1, C_2, C_3, C_4) exclusively establishes an algebraic $(1,3)$ Minkowski signature.*

Proof. Extracting the strict 4×4 distance matrix for the array $[C_1, C_2, C_3, C_4]$ and evaluating the characteristic polynomial precisely factors into the following four exact algebraic eigenvalues:

$$\begin{aligned} \lambda_0 &= +(3+\sqrt{3})r \qquad &&\text{(Time-like dimension)} \\ \lambda_1 &= (1-\sqrt{3})r \qquad &&\text{(Space-like dimension 1)} \\ \lambda_2 &= -(3-\sqrt{3})r \qquad &&\text{(Space-like dimension 2)} \\ \lambda_3 &= -(1+\sqrt{3})r \qquad &&\text{(Space-like dimension 3)} \end{aligned}$$

This perfect $(1,3)$ signature mirrors the $(+,-,-,-)$ metric of physical spacetime. Furthermore, this pure distance metric geometry yields two exact structural constants:

1. **Invariant Volume Limit:** The structural metric determinant yields $\det(g) = \prod \lambda_i = -12r^4$, establishing an internal volume invariant without arbitrarily tuned parameters.
2. **Dimensionless eigen-ratio:** The ratio of the distinguished positive and negative eigenvalue scales evaluates exactly to

$$\frac{|\lambda_0|}{|\lambda_3|} = \frac{3+\sqrt{3}}{1+\sqrt{3}} = \frac{(3+\sqrt{3})(\sqrt{3}-1)}{(\sqrt{3}+1)(\sqrt{3}-1)} = \frac{2\sqrt{3}}{2} = \sqrt{3}$$

This defines a dimensionless geometric invariant of the certified distance matrix. Its identification with the physical speed of light would require an additional unit-matching correspondence.

$\square$

Geometric Centrality and the Origin of Mass (Mach's Principle):

In a complete discrete metric space, the inertial coupling of a node i to the full graph is measured by an L^2 Machian potential—the sum of *squared* normalized distances to all other nodes:

$$V_i = \sum_{j \neq i} \left(\frac{D_{ij}}{r} \right)^2 .$$

This is the natural discrete analogue of a quadratic mass term $\frac{1}{2}\phi^T \mathbf{M} \phi$ on the lattice (Section 4): it stays in $\mathbb{Q}(\sqrt{3})$ without invoking $\sqrt{13}$ in the mass ordering proofs, and is machine-checked in the companion Lean file (`centralityPotentialSq`). A lower V_i indicates deeper centrality (heavier effective mass). Computing this for the 17 nodes reveals a hierarchy matching physical mass dynamics:

1. **Rank 1 (A, $V \approx 21.75$ in units of r^2):** The absolute structural origin.
2. **Rank 2 (X_{17}, $V \approx 26.07$):** The unique degree-4 node on the parity-symmetric axis constitutes the second-deepest potential well.
3. **Sub-orbit Divergence:** The nodes embedded natively in $\mathbb{Q}(\sqrt{3})$ (C_1, C_3) have $V \approx 30.32$, locating them significantly deeper in the central potential well than the dimensionally equivalent nodes (C_2, C_4) which require the algebraic extension $\mathbb{Q}(\sqrt{13})$ at $V \approx 47.18$.

Geometric centrality (V_i) functions as a purely topological measure of a node's inertial coupling relative to the entire structure.

3 Node Classification by Geometric Invariants

The 17 nodes of the graph are classified by two rigid, Lean-verified geometric invariants: their **degree** (connectivity) and their **algebraic field membership** (whether their metric coupling to X_{17} resolves in $\mathbb{Q}(\sqrt{3})$ or requires the extension $\mathbb{Q}(\sqrt{13})$). No physical labels are assigned at this stage.

1. **The Origin (A):** The unique degree-5 node, maximally connected.

2. **The Axis Node (X_{17}):** A degree-4 node lying exactly on the parity-symmetric x-axis ($y = 0$), invariant under the $\mathbb{Z}/2$ mirror automorphism.

3. **Detachable Leaves (B, P_4, P_6):** Three degree-1 nodes on the outer boundary.

4. **Confined Boundary Chains ($K, L, M, N,$ Top, Bot, P_3, P_5):** Eight degree-2 nodes forming connected chain pathways.

5. **$\mathbb{Q}(\sqrt{3})$-coupled Internal Nodes (C_1, C_3):** Degree-4 nodes whose squared distance to X_{17} equals $r^2/4 \in \mathbb{Q}$. They participate in the right triangles Δ_1 and Δ_3.

6. **$\mathbb{Q}(\sqrt{13})$-coupled Internal Nodes (C_2, C_4):** Degree-3 nodes whose squared distance to X_{17} equals $13r^2/4$, requiring the algebraic extension $\mathbb{Q}(\sqrt{13})$. They participate in the equilateral triangle Δ_2.

The algebraic field boundary splitting items 5 and 6 is a Lean-verified theorem (Theorem 5). The degree sequence is likewise Lean-verified (Theorem 4). This classification is forced by the geometry, not chosen by the author.

4 The Discrete Geometric Lagrangian

If this rigid geometry structures dynamic fields, it must formally map onto a discrete scalar topological action. In a discrete causal lattice, space does not exist continuously between points; the continuous spatial gradient operator ∇^2 must natively map to its discrete spectral equivalent, the Graph Laplacian $\Delta = \mathbf{D} - \mathbf{A}$.

Applying standard Lattice Field Theory mathematics to the complete 17-point Generation 1 topology yields a natural parameter-free discrete scalar action associated with the certified topology:

$$S = \frac{1}{2}\vec{\phi}^T \Delta \vec{\phi} - \frac{1}{2}\vec{\phi}^T \mathbf{M} \vec{\phi} - \frac{\alpha}{3!} \sum_{i,j,k \in \text{Triangles}} \phi_i \phi_j \phi_k \tag{4}$$

Here, each component relies strictly on derived geometric operators:

1. **Kinetic Operator (Δ):** The kinematic propagation of the ϕ field is strictly constrained by the 21 atomic lines forming the symmetric 17×17 Graph Laplacian. Numerical extraction proves the nullity of this matrix is exactly 1, guaranteeing a unique, translation-invariant degenerate vacuum ground state ($\Delta \vec{1} = \vec{0}$).

2. **Mass Matrix (M):** The topological centrality of a node dictates its L^2 coupling. We construct the strictly diagonal matrix $\mathbf{M} = \text{diag}(V_i)$ where $V_i = \sum_j (D_{ij}/r)^2$. The symmetric origin possesses the lowest scalar coupling; boundary and $\mathbb{Q}(\sqrt{13})$ nodes carry the largest V_i.

3. **Interaction Vertices:** A 3-point fundamental interaction (ϕ^3) maps topologically to closed 3-cycles. Computational extraction identifies exactly three fundamental coupling triangles (Δ_1: $A - C_2 - C_4$, Δ_2: $A - C_1 - X_{17}$, Δ_3: $A - C_3 - X_{17}$).

4.1 Analytical Generational Scale Ratios

We can exploit the Discrete Lagrangian directly to derive the tree-level base scalar hierarchies of the three triangles without internal fitting parameters.

Reducing the full 17-component positional vector field $\vec{\phi}$ into effective domain macrostates natively corresponding to the three triangles ($\phi_1 \leftrightarrow \Delta_1, \phi_2 \leftrightarrow \Delta_2, \phi_3 \leftrightarrow \Delta_3$), we can evaluate the effective scalar potential. By isolating the quadratic term $\frac{1}{2}\vec{\phi}^T \mathbf{M} \vec{\phi}$, we integrate over the intrinsic geometric potentials of the anchor nodes.

Generation I (Δ_1: $A - C_2 - C_4$) relies exclusively on the spatial origin A. In contrast, Generations II and III (Δ_2, Δ_3) critically depend on X_{17} to geometrically seal their discrete structural paths.

Extracting from the L^2 potentials at unit radius:

- $V_A = \sum_j (D_{A,j}/r)^2 \approx 21.75$
- $V_{X_{17}} = \sum_j (D_{X_{17},j}/r)^2 \approx 26.07$

Under this effective reduction, one obtains deterministic tree-level scalar scales from geometric centrality:

$$m_{\text{I}}^2 \propto V_A \approx 21.75$$
$$m_{\text{II}}^2 = m_{\text{III}}^2 \propto V_A + V_{X_{17}} \approx 47.82$$

The predicted baseline ratio is:

$$\frac{m_{\text{II,III}}}{m_{\text{I}}} = \sqrt{\frac{V_A + V_{X_{17}}}{V_A}} \approx 1.48$$

Part II

Physical Isomorphisms (Derivations)

5 Physical Isomorphisms from a Certified Discrete Kernel

The structural correspondences proposed in this paper arise from a hybrid methodology. First, the 17-point Vesica Piscis construction is encoded and certified in Lean 4 at the level of exact coordinates, graph invariants, algebraic field obstructions, simplicial boundary relations, and selected spectral quantities. Second, standard analytical mathematics—representation theory, spectral graph theory, discrete differential geometry, and continuum analogies from mathematical physics—is applied to interpret those certified invariants.

The resulting physical sections should therefore be read with the following distinction in mind:

- **Lean-verified facts:** exact discrete statements about the finite geometric/combinatorial structure.
- **Analytical derivations:** mathematically structured interpretations built on top of that certified discrete substrate.
- **Open bridges:** statements whose rigorous derivation would require additional constructions not yet carried out here (for example, a fully defined discrete Dirac operator with an index theorem, or a proved continuum-limit convergence theorem).

In this sense, the 17-point geometry is not claimed to *already contain* the complete continuous Standard Model as a formal theorem. Rather, it provides an unusually rigid zero-parameter discrete scaffold whose verified invariants exhibit nontrivial correspondences with prominent structural features of particle physics and gravitation.

5.1 Comprehensive Standard Model Accounting

The structural correspondences between the 17-point geometry and the Standard Model arise from applying standard analytical physics to the Lean-verified discrete topological baseline. The 17-point algorithm does not generate 61 individual nodes for the 61 physical elementary particles. Instead, it generates the exact mathematical generators and dimensions required to produce the Standard Model architecture. The table below summarizes these correspondences, distinguishing the Lean-verified geometric premise from the analytical physical derivation.
In each row, the third column is a Lean-verified fact about the discrete graph; the fourth column is the analytical physical derivation applied to that fact using standard representation theory, spectral graph theory, or discrete differential geometry.

5.2 Parity Violation and Chiral Asymmetry

The construction has an intrinsic left–right asymmetry: the node $X_{17} = (50\sqrt{3}, 0)$ exists on the $+x$ side of the origin A, with no mirror node at $(-50\sqrt{3}, 0)$ (Theorem 3). This geo-

Physics Component	**Standard Model**	**Lean-Verified Geometric Premise**	**Analytical Derivation**
Gluons	$SU(3)_C$ (8 states)	8 degree-2 boundary nodes (Lean: Thm 4)	Adjoint rep dimension: $N^2 - 1 = 8 \implies \mathfrak{su}(3)$.
Weak Bosons	$SU(2)_L$ (3 states)	3 degree-1 leaf nodes (Lean: Thm 4)	Adjoint rep dimension: $N^2 - 1 = 3 \implies \mathfrak{su}(2)$.
Photon	$U(1)_Y$	Single degree-5 origin node A (Lean: Thm 4)	Central $U(1)$ symmetry locus.
Higgs VEV	$T_3 = 0$ SSB locus	X_{17} lies on axis, fixed under mirror parity (Lean-verified)	VEV interpretation of X_{17} magnitude $v = r\sqrt{3}/2$.
Generations	3 (e/μ/τ)	Exactly 3 triangles; computationally certified candidate 2-simplex sector	Analytical identification with a discrete topological index suggests a 3-generation sector; a rigorous Dirac/index derivation remains open.
Quarks	Color-charged fermions	C_1, C_3: $d^2 = r^2/4 \in \mathbb{Q}$ (Lean: Thm 5)	Native $\mathbb{Q}(\sqrt{3})$ coupling to the boundary sector suggests a color-coupled interpretation.
Leptons	Colorless fermions	C_2, C_4: $d^2 = 13r^2/4$ (Lean: Thm 5)	$\mathbb{Q}(\sqrt{13})$ separation suggests a color-singlet interpretation under the stated projection mechanism.
Gravity	Curvature	$S_{\mathrm{FR}} = -15$ (Lean-verified)	Discrete curvature invariant; continuum-limit convergence to Einstein-Hilbert remains open.

metric chirality permanently breaks left–right symmetry for all X_{17}-coupled interactions, suggesting a structural origin for parity violation in the weak interaction [8].

5.3 Gravitational Dynamics and Discrete Curvature

Lean-verified premise: The Forman-Ricci curvature $\mathbf{F}(u, v) = 4 - \deg(u) - \deg(v) + 3|\Delta_{u,v}|$ is computed exactly over all 21 edges. The Lean kernel evaluates the total discrete action $S = \sum \mathbf{F}(u, v) = -15$. This is a topological invariant of the Gen 1 graph—analogous to how every cube has Euler characteristic $V - E + F = 2$.
Analytical derivation: In discrete differential geometry, the Forman-Ricci scalar curvature is the standard combinatorial analogue of the Riemann curvature tensor restricted to 1-simplices. The negative global curvature ($S = -15$) certifies nontrivial discrete curvature in the chosen Forman-Ricci sense. We conjecture that, by analogy with established convergence results for Regge calculus on triangulated manifolds, as recursion depth $n \to \infty$ and the mesh density increases, sums of discrete Forman-Ricci curvatures on this recursive construction could converge to the continuous Einstein-Hilbert action $\int R\sqrt{-g}\, d^4x$. Establishing this convergence rigorously for the specific fractal-like recursive geometry considered here remains an open problem. The discrete geometry thus provides a native, parameter-free geometric framework for gravitational curvature.

5.4 Generational Recursion as the Engine of Dynamics

The Standard Model and General Relativity are continuous dynamic theories. In this framework, dynamics (movement, expansion, and time evolution) are not governed by an external background clock, but are mathematically identical to the unceasing recursive execution of the algorithm itself ($Gen_n \to Gen_{n+1}$).

Theorem 10 (Recursive Scale Consistency)**.** *Let the algorithm's recursive step be defined as the operator $\mathcal{T}$, mapping the geometric state G_n to G_{n+1} by converting every atomic segment of G_n into the fundamental radius of a new localized Vesica Piscis state.*

Proof: *The Lean formalization certifies that the recursive ansatz employs an explicitly computable scale factor in $\mathbb{Q}(\sqrt{3})$, and that its first iterates are positive and bounded above by the seed radius in the cases checked. Interpreting repeated application of this scale-consistent recursive ansatz as a directed dynamical process is analytical.*

Analytical interpretation: In continuous physics, dynamics are defined by a time-evolution operator $U(t) = e^{-iHt}$, and scale-dependencies are governed by the Renormalization Group (RG) flow. In this algorithmic baseline, the recursive operator $\mathcal{T}$ suggests an interpretation unifying both:

1. **Cosmological Expansion:** Because every atomic segment spawns a new spatial generation, the number of nodes and the effective volume of the graph grow in the idealized non-overlap counting model, providing a candidate parameter-free geometric origin for spatial expansion.

2. **The Continuum Limit (Dynamics):** As recursion depth $n \to \infty$, the discrete segments become arbitrarily dense. The stepwise application of $\mathcal{T}$ suggests a route to continuous dynamic evolution, allowing the discrete action evaluated by the graph Laplacian at finite n (Section 4) to potentially converge into the continuous Lagrangians of Quantum Field Theory.

Consequently, the 17-point graph is not a static lattice; it is the $n = 1$ initial condition of an infinite recursive operator that motivates an interpretation in which causal physical dynamics emerges from recursive refinement.

5.5 Absolute Topological Bounds (UV/IR Cutoffs)

The initial checked recursive scaling iterates are strictly positive and bounded above by the seed radius. This motivates the conjecture that all recursively generated atomic lengths satisfy $0 < l \le r$, though a full global proof remains open.

Let L be the set of all atomic segment lengths generated as $n \to \infty$. The Lean formalization verifies positivity and boundedness for the initial checked scaling iterates. A full theorem asserting $\forall l \in L, \quad 0 < l \le r$ for all recursively generated atomic lengths remains open.

Physical derivation (The Absence of Singularities): In continuous Quantum Field Theory and General Relativity, interactions exhibit catastrophic divergences as distance $d \to 0$ (ultraviolet) or as scale $d \to \infty$ (infrared). In this algorithmic framework, the initial checked recursive scaling iterates remain strictly positive and below the seed radius, suggesting a possible **Ultraviolet (UV) cutoff** (a strictly non-zero minimum topological distance, functioning as an intrinsic Planck-scale bound) and an **Infrared (IR) cutoff** (the maximum structural radius r, functioning as a cosmological horizon). A full global proof that zero distances never arise and that all recursively generated atomic lengths remain bounded above by r is left open.

5.6 A Field-Theoretic Split Suggestive of the Quark/Lepton Distinction

Lean-verified premise: By Theorem 5, the squared distances from the internal nodes to the symmetry-breaking origin X_{17} fracture into two distinct algebraic classes: $\{C_1, C_3\}$ couple natively at $d^2 = r^2/4 \in \mathbb{Q}$, while $\{C_2, C_4\}$ couple at $d^2 = 13r^2/4$. The Lean 4 kernel further verifies that $\sqrt{13/4}$ has no representation in the $\mathbb{Q}(\sqrt{3})$ arithmetic model.

Analytical derivation: The strong-force boundary sector of the graph is generated entirely within the native coordinate field $\mathbb{F} = \mathbb{Q}(\sqrt{3})$. It is therefore natural to ask whether internal nodes whose linear metric couplings resolve natively in $\mathbb{F}$ behave differently from those whose couplings require an orthogonal algebraic extension.

1. For nodes $\{C_1, C_3\}$, the linear amplitude evaluates natively to $r/2 \in \mathbb{Q}(\sqrt{3})$. These amplitudes remain closed over the base field and therefore admit direct interaction with the native boundary sector.

2. For nodes $\{C_2, C_4\}$, the linear amplitude requires the extension $\mathbb{Q}(\sqrt{13})$. A natural projection of such an extension-valued quantity back to the base field is given by the Galois trace. Under that projection, the purely $\sqrt{13}$ contribution cancels:

$$\mathrm{Tr}_{\mathbb{Q}(\sqrt{3},\sqrt{13})/\mathbb{Q}(\sqrt{3})}\left(\frac{r\sqrt{13}}{2}\right) = \frac{r\sqrt{13}}{2} + \left(-\frac{r\sqrt{13}}{2}\right) = 0.$$

This does *not* by itself prove a physical quark/lepton ontology. It does, however, provide a mathematically explicit mechanism by which one internal pair remains visible to the native $\mathbb{Q}(\sqrt{3})$ boundary sector while the other pair is annihilated under projection to that field. Under the stated physical interpretation, this motivates identifying $\{C_1, C_3\}$ with a color-coupled sector and $\{C_2, C_4\}$ with a color-singlet sector. In that sense, the $\mathbb{Q}(\sqrt{13})$ obstruction provides a structured candidate mechanism for the quark/lepton split and for neutrino-mass suppression analogies.

5.7 Emergent $(1, 3)$ Lorentzian Spacetime

Lean-verified premise: The 4×4 distance sub-matrix restricted to $\{C_1, C_2, C_3, C_4\}$ has four exact eigenvalues with signature $(+, -, -, -)$ (Theorem 6, Lean-verified).
Analytical derivation: This perfect $(1, 3)$ Minkowski signature yields a dimensionless geometric eigen-ratio:

$$\frac{|\lambda_0|}{|\lambda_3|} = \frac{3+\sqrt{3}}{1+\sqrt{3}} = \sqrt{3}$$

This ratio defines a dimensionless geometric eigen-bound of the Lean-verified distance matrix; its identification with the physical speed of light c would require a unit-matching correspondence between the atomic segment length and Planck-scale physics.

6 Spectral Derivations from the Certified Symmetries

The following three derivations apply standard analytical physics to the Lean-verified discrete invariants of Part I.

6.1 The Geometric Origin of Weak Isospin (T_3) and the $10 \oplus 7$ Split

Lean-verified premise: The graph automorphism group is exactly $\mathbb{Z}/2$, consisting solely of the identity and a spatial parity mirror reflection $\mathbf{R} : y \to -y$ (Lean-verified). The three nodes on the x-axis (A, B, X_{17}) are fixed under $\mathbf{R}$.
Formal Analytical Derivation: Let $V \cong \mathbb{R}^{17}$ be the vector space spanned by the 17 nodes. Because $\mathbf{R}^2 = I$, the space decomposes into ± 1 eigenspaces $V = V_+ \oplus V_-$. The Lean 4 kernel formally verified (Theorem `parity_eigenspaces_10_7`) that the trace count induced by the mirror permutation on the 17-node basis is exactly 3 (the number of fixed axis nodes). This rigorous geometric constraint forces the exact eigenspace dimensions $\dim(V_+) = 10$ and $\dim(V_-) = 7$.

Defining the weak isospin operator $\hat{T}_3$ as the projection onto the antisymmetric subspace V_-, the node X_{17} lies entirely in V_+ and has exactly zero weak isospin. This makes X_{17} the natural candidate for the Spontaneous Symmetry Breaking (SSB) locus: it possesses a non-zero, parity-invariant scalar magnitude ($v = r\sqrt{3}/2$), analytically paralleling the vacuum expectation value required by the Goldstone theorem to break the electroweak symmetry and generate mass.

6.2 Cartan's Classification and Gauge-Algebra Correspondence

Lean-verified premise: The degree sequence (Theorem 4, Lean-verified) partitions the outer boundary into exactly two disconnected topological subsets:

1. Degree-1 leaves: $S_1 = \{B, P_4, P_6\}$, so $|S_1| = 3$.

2. Degree-2 confined chains: $S_2 = \{K, L, M, N, \mathrm{Top}, \mathrm{Bot}, P_3, P_5\}$, so $|S_2| = 8$.

Because S_1 and S_2 share no edges, they form topologically distinct boundary sectors.

Analytical derivation: In the Standard Model, the weak and strong gauge bosons transform in the adjoint representations of compact simple Lie algebras. By Cartan's classification:

- the unique compact simple Lie algebra with adjoint dimension 3 is $\mathfrak{su}(2)$;
- the unique compact simple Lie algebra with adjoint dimension 8 is $\mathfrak{su}(3)$.

This does *not* yet prove that the graph itself carries a fully realized gauge theory with explicit structure constants, local connection data, and Yang–Mills dynamics. What it does show is that the Lean-verified boundary dimensionalities align exactly with the adjoint dimensions required for the weak and strong gauge algebras, and do so in a topologically separated way. Accordingly, the geometry provides a mathematically rigid correspondence with the $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$ sectors of the Standard Model.

A full derivation of gauge-algebra brackets from graph holonomies or discrete connection data remains an important open problem.

6.3 Discrete Exterior Calculus and a Candidate Three-Generation Sector

Lean-verified premise: The graph admits exactly three 2-simplices (triangles). We explicitly encode discrete boundary operators

$$B_1 \in \mathbb{Z}^{17\times 21}, \qquad B_2 \in \mathbb{Z}^{21\times 3},$$

mapping edges to nodes and triangles to edges, respectively. The Lean 4 kernel verifies the simplicial identity

$$B_1 B_2 = 0,$$

thereby certifying that the chosen finite structure carries a consistent boundary-of-boundary-zero relation. We further construct the encoded 2-sector Hodge Laplacian

$$\Delta_2 = B_2^T B_2,$$

and Lean verifies that its determinant is nonzero (in the present encoding, $\det(\Delta_2) = 24$), so the encoded 2-simplex sector has full rank and dimension 3.

Analytical derivation: In discrete differential geometry and spectral formulations of fermionic matter, the topology of higher-degree form sectors can control the multiplicity of physically relevant modes. The present geometry therefore exhibits a particularly suggestive feature: it carries an explicitly certified 3-dimensional 2-simplex/2-form sector.

This is not yet a proof of three fermion generations. To obtain that, one would need:

1. an explicit discrete Dirac or Kähler–Dirac operator on the certified complex,
2. a rigorous spectral analysis of its chiral zero-modes,
3. and a corresponding discrete index theorem relating those modes to the topological sector isolated above.

Accordingly, the mathematically correct conclusion at this stage is more modest but still nontrivial: the Lean-certified topology supports a natural *candidate* three-generation sector. Establishing the exact link between this sector and observed fermion generations is the central open problem connecting the discrete geometry to Standard Model matter content.

6.4 Recursive Self-Similarity and the Continuum-Gravity Conjecture

A recurrent issue in discrete approaches to spacetime is how, or whether, a continuous gravitational theory emerges in an appropriate large-scale limit. In the present framework, the central certified inputs are:

- a nonzero discrete curvature invariant $S_{\mathrm{FR}} = -15$,
- an exactly computable recursive scale factor in $\mathbb{Q}(\sqrt{3})$,
- and a self-similar generation rule in which atomic segments seed new local copies of the construction.

Analytical conjecture: These features suggest a possible scale-consistent refinement process in which the recursive geometry behaves like a discretely self-similar curved complex. By analogy with Regge-calculus and simplicial-gravity limits, one may conjecture that sufficiently deep recursive refinement could admit a continuum description in which averaged discrete curvature observables converge toward an Einstein–Hilbert-type action.

At present, however, this remains a conjectural bridge rather than a theorem. In particular, the following have *not* yet been proved:

1. existence of a renormalization-group fixed point for the recursive geometry,
2. diffeomorphism invariance of a continuum limit,
3. convergence of the discrete curvature observables to the Einstein-Hilbert action,
4. or emergence of Einstein's equations.

The mathematically justified claim is therefore narrower: the certified geometry supplies a rigid recursive curved scaffold that makes a continuum-gravity interpretation plausible, while the exact continuum-limit theorem remains open.

7 Algorithmic Invariance and Falsifiability

Because the Vesica Piscis construction relies on zero free parameters, this framework cannot be "tuned" to accommodate conflicting experimental data. The geometry represents an absolute, rigid mathematical scaffold. Consequently, the physical derivations are strictly vulnerable to binary falsification.

The proposed physical identification of the certified geometry would be decisively challenged if experimental physics confirms any of the following constraints:

1. **A Fourth Fermion Generation:** The certified geometry contains exactly three triangles and supports, under the chosen edge-incidence encoding, a candidate 3-generator 2-simplex sector (Theorem `second_betti_sector_dimension_three`). A rigorous derivation of exact fermion-generation counting from a discrete Dirac/index theorem remains open (§6.3); once established, discovery of a fourth generation would falsify that identification.

2. **Non-Standard Gauge Structures:** The Lean-verified boundary degree sequence generates exactly 1, 3, and 8 isolated degrees of freedom. Discovery of fundamental forces requiring $SU(4)$ or $SU(5)$ at baseline would falsify the graph.

3. **Deviation from (1,3) Lorentzian Signature:** The Lean-verified spacetime submatrix restricts to a $(1,3)$ Lorentzian signature (Theorem 6). Empirical confirmation of fundamentally continuous higher-dimensional spacetime would conflict permanently with this bound.

If none of these falsification conditions are met, the Vesica Piscis algorithmic scaffold, to our knowledge, remains an unusual zero-parameter mathematical structure exhibiting this combination of certified invariants capable of analytically corresponding to the Standard Model's dimensional and algebraic architecture from compass-and-straightedge first principles.

8 Conclusion

This paper develops a rigorous discrete-geometric analysis of the 17-point Vesica Piscis construction using a hybrid methodology. The Lean 4 theorem prover serves as a computational certificate for the foundational finite structure: exact coordinates in $\mathbb{Q}(\sqrt{3})$, a 21-edge atomic graph, exactly three triangles, a strict $\mathbb{Z}/2$ automorphism group, a rigid $\mathbb{Q}(\sqrt{13})$ algebraic field boundary, an emergent $(1,3)$ eigenvalue signature on the distinguished spacetime submatrix, and a nonzero Forman-Ricci curvature invariant.

On top of that certified substrate, standard analytical mathematics—including representation theory, spectral graph theory, and discrete differential geometry—reveals a collection of nontrivial structural correspondences with the architecture of the Standard Model. In particular, the geometry isolates boundary sectors of dimensions 3 and 8, exhibits a parity-induced $10 \oplus 7$ basis decomposition, and supports an explicitly encoded candidate 3-generator 2-simplex sector. These facts do not yet constitute a full derivation of gauge theory, fermion generations, or gravitation, but they provide a mathematically rigid candidate scaffold from which such structures may plausibly emerge.

Several central bridges remain open. These include:

1. construction of an explicit discrete Dirac/Kähler–Dirac operator and proof of an index theorem,

2. derivation of Lie-algebra brackets or discrete connection data yielding gauge dynamics,

3. and a rigorous continuum-limit theorem connecting the recursive discrete geometry to continuous gravitational field theory.

Accordingly, the strongest justified conclusion at present is not that the Standard Model has been fully derived from compass-and-straightedge geometry, but that the 17-point Vesica Piscis seed provides an unusually rigid zero-parameter discrete foundation exhibiting several exact invariants strikingly aligned with core structural features of known physics. Whether those correspondences can be elevated into a complete unified derivation remains an important open program.

References

[1] Euclid, *Elements*, c. 300 BCE. Translation: T. L. Heath, *The Thirteen Books of Euclid's Elements*, Cambridge University Press, 1908.

[2] R. L. Workman *et al.* (Particle Data Group), "Review of Particle Physics," *Prog. Theor. Exp. Phys.* **2022**, 083C01 (2022), updated 2024.

[3] C. F. Gauss, *Disquisitiones Arithmeticae*, Leipzig, 1801.

[4] S. Weinberg, "A Model of Leptons," *Phys. Rev. Lett.* **19**, 1264 (1967).

[5] P. W. Higgs, "Broken Symmetries and the Masses of Gauge Bosons," *Phys. Rev. Lett.* **13**, 508 (1964).

[6] C. N. Yang and R. L. Mills, "Conservation of Isotopic Spin and Isotopic Gauge Invariance," *Phys. Rev.* **96**, 191 (1954).

[7] M. Gell-Mann, "Symmetries of Baryons and Mesons," *Phys. Rev.* **125**, 1067 (1962).

[8] C. S. Wu *et al.*, "Experimental Test of Parity Conservation in Beta Decay," *Phys. Rev.* **105**, 1413 (1957).

[9] F. Lindemann, "Über die Zahl π," *Math. Ann.* **20**, 213–225 (1882).